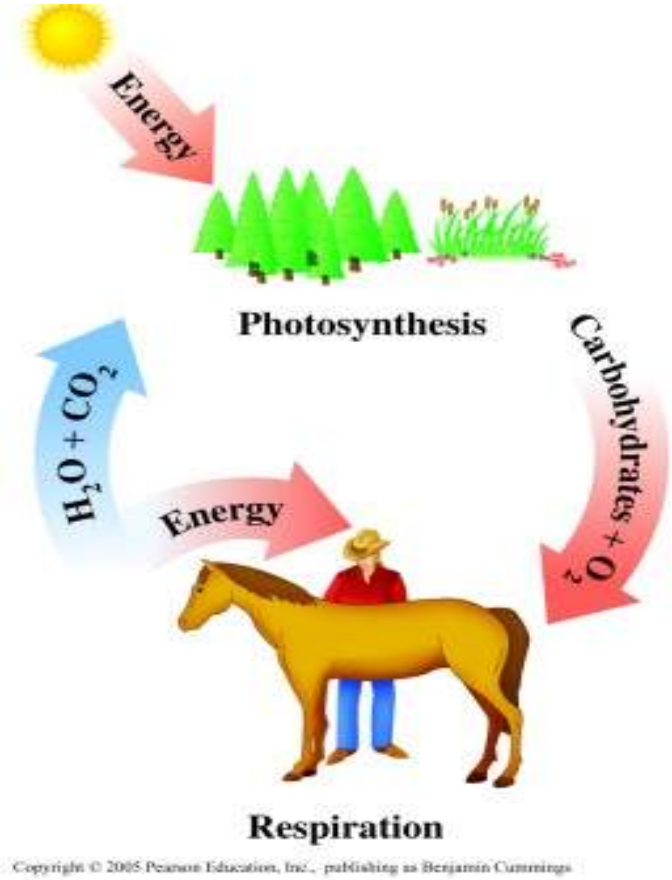
The Biochemical Importance of Carbohydrates

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Carbohydrates

- The term Carbohydrates is applied to a large number of relatively heterogeneous compounds found in all animals and plants.
- They are most abundant biomolecules on earth.



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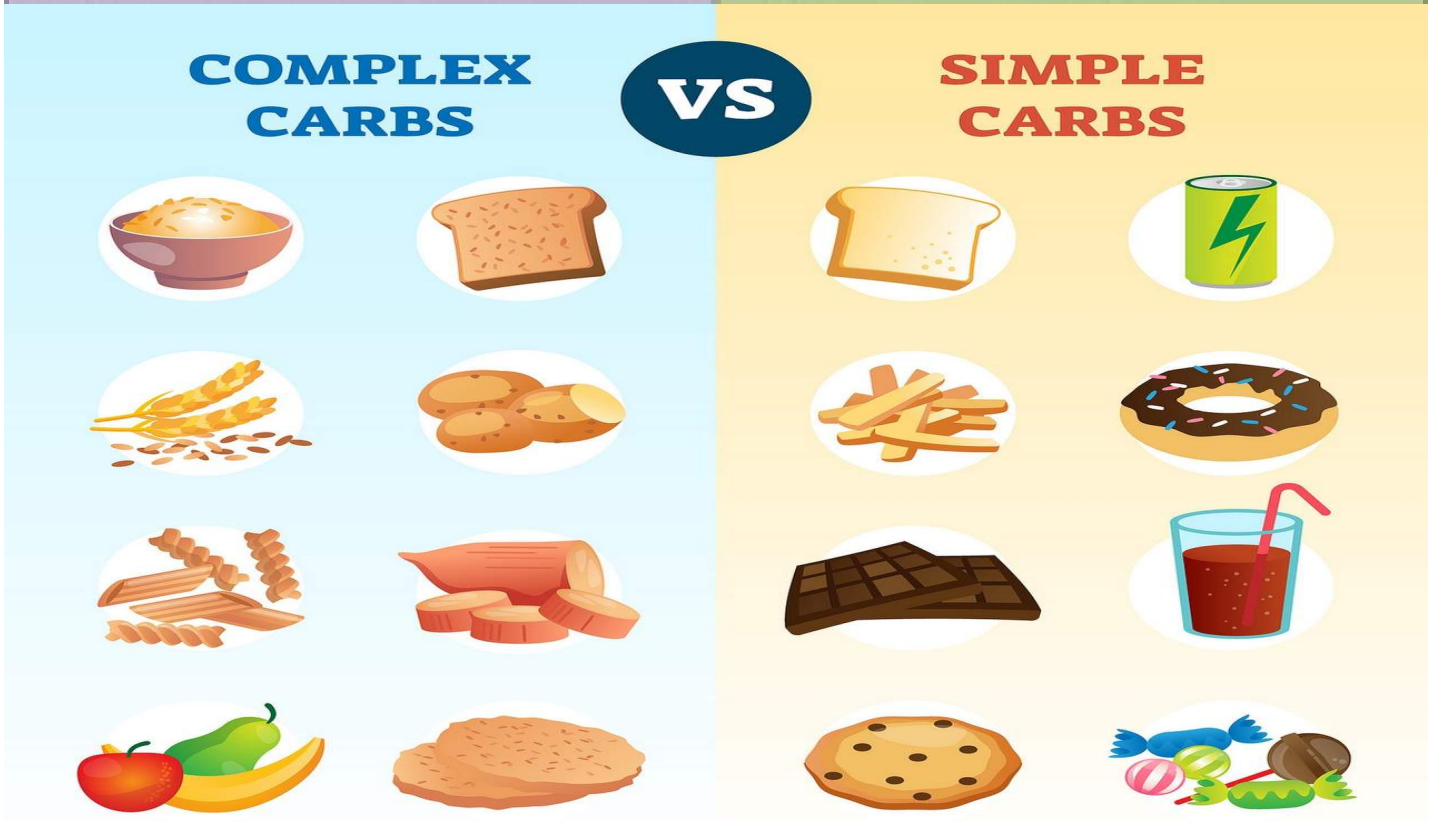
Biochemical role of Carbohydrates

- They have a wide range of functions, including:
  - Providing a significant fraction of the dietary calories for most organisms.
  - Acting as a storage form of energy in the body.
  - Serving as cell membrane components.
  - Serve as a structural component of many organisms, including the cell walls of bacteria, the exoskeleton of many insects, and the fibrous cellulose of plants.
  - To provide the cells with carbon atoms for building molecules of protein, lipids, and nucleic acids.

Main carbohydrates of diet of humans

- According to the Dietary Guidelines, carbohydrates should make up 45-65 percent of our daily calories.
- One gram of carbohydrates provide about 4 calories of energy. They have the advantage of being converted into energy faster than fats and protein.
- Each day carbohydrates are utilized in foods such as bread, pasta, potatoes, and rice.
- Other carbohydrates called disaccharides include sucrose (table sugar) and lactose in milk.
- During digestion and cellular metabolism, carbohydrates are converted into glucose, which is oxidized further in cells to provide energy.

Complex Carbohydrates	Simple Carbohydrates
Made of long chains of sugar molecules	Made of short chains of sugar molecules
Takes time to digest	Easy to digest
Not always very sweet	Sweet
Slowly increases blood sugar levels	Rapidly increases blood sugar level



Instant source of energy

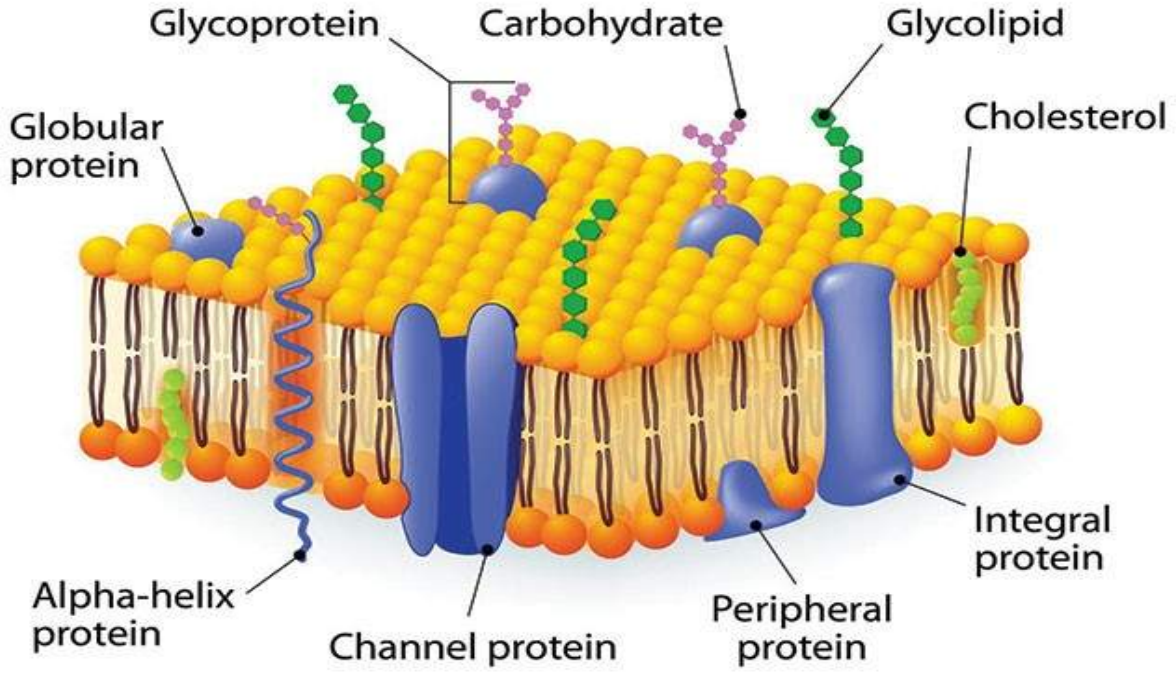
- Carbohydrate are the first food components which get digested in body giving out energy.
- Carbohydrates are generally available as an immediate energy source, whereas lipids act as a long-term energy resource and tend to be utilized at a slower rate.
- A single molecule of glucose produces 38 molecules of ATPs.
- Tissues, such as the brain, red blood cells (RBCs), kidney medulla, and exercising muscle, require a continuous supply of glucose as a metabolic fuel.
- A constant source of blood glucose is an absolute requirement for human life.
- The proper regulation of glucose metabolism is of paramount importance to survival.

Storage form

- Starch is the chief storage form of carbohydrates in chlorophyll-containing plants.
- The energy stored by plants during the formation of carbohydrates is used to carry out mechanical work and to perform biosynthetic activities.
- Carbohydrates are stored in human body in the form of glycogen. Glycogen is a multibranched polysaccharide of glucose that serves as a form of energy storage. Glycogen primarily stored in the cells of the liver and muscles.
- During starvation, glycogen act as primary source of energy by providing ATPs.

Cell membrane component

- Carbohydrates have important structural roles.
- Carbohydrates are attached to proteins and lipids on the exterior side of cell membrane.



Glycoproteins

- Glycoproteins are one class of glycoconjugate or complex carbohydrate —molecules containing one or more carbohydrate chains covalently linked to protein (to form glycoproteins or proteoglycans) or lipid (to form glycolipids).
- Many cell membrane proteins contain substantial amounts of carbohydrate.
- Most of the plasma proteins, and many peptide hormones, are glycoproteins.
- The human blood groups A, B, AB, and O depend on the oligosaccharide part of the glycoprotein or a glycolipid on the surface of erythrocytic cells

Functions Served by Glycoproteins

Transport molecule	Transferrin, ceruloplasmin
Immunological molecule	Immunoglobulins, histocompatibility antigens
Hormone	Chorionic gonadotropin, thyroid-stimulating hormone (TSH)
Enzyme	Various, eg, alkaline phosphatase
Cell attachment–recognition site	Various proteins involved in cell–cell (eg, sperm-oocyte), virus-cell, bacterium-cell, and hormone-cell interactions
Interact with specific carbohydrates	Lectins, selectins (cell adhesion lectins), antibodies
Receptor	Cell surface proteins involved in hormone and drug action
Hemostasis (and thrombosis)	Specific glycoproteins on the surface membranes of platelets

Glycosaminoglycans (GAGs) and proteoglycans

- Most of the GAGs are covalently conjugated to a protein core, the product of which is termed proteoglycans.
- They are formed mainly of carbohydrates (95%) and only (5%) proteins.
- GAGs are present mainly in the extracellular matrix (ECM) or ground substance in association with other extracellular proteins.

Functions of GAGs and proteoglycans

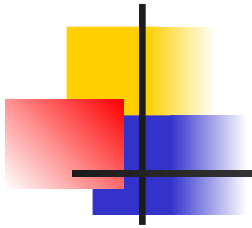
- They are important constituents of extracellular matrix.
- Provides flexible mechanical support for the ECM.
- Acts as a lubricant in synovial fluid.
- GAGs solutions act as shock absorbents
- The proteoglycans interact with a variety of proteins in the matrix, such as collagen and elastin, and theses interactions are important in determining the structural organization of the matrix.
- Hyaluronic acid proteoglycans involved in wound healing (repair). In the initial phase of wound healing, hyaluronic acid concentration increases many folds at the wound site.

Pentoses of Physiological Importance

Sugar	Source	Biochemical and Clinical Importance
D-Ribose	Nucleic acids and metabolic intermediate	Structural component of nucleic acids and coenzymes, including ATP, NAD(P), and flavin coenzymes
D-Ribulose	Metabolic intermediate	Intermediate in the pentose phosphate pathway
D-Arabinose	Plant gums	Constituent of glycoproteins
D-Xylose	Plant gums, proteoglycans, glycosaminoglycans	Constituent of glycoproteins

Hexoses of Physiological Importance

Sugar	Source	Biochemical Importance
D-Glucose	Fruit juices, hydrolysis of starch, cane or beet sugar, maltose and lactose	The main metabolic fuel for tissues; "blood sugar"
D-Fructose	Fruit juices, honey, hydrolysis of cane or beet sugar and inulin, enzymic isomerization of glucose syrups for food manufacture	Readily metabolized either via glucose or directly
D-Galactose	Hydrolysis of lactose	Readily metabolized to glucose; synthesized in the mammary gland for synthesis of lactose in milk. A constituent of glycolipids and glycoproteins
D-Mannose	Hydrolysis of plant mannan gums	Constituent of glycoproteins



Thank you